

# GRID CONTROL — Build Plan (Jun 18 2026)

Safe word: **GRIDLOCK-RESUME-18JUNE**. Supersedes [GRIDLOCK-RESUME-16JUN]. Master sequence (Gaurav, unchanged): **Backend → Frontend → Stitch → Test brand → Launch**. This plan finishes step 1 (backend) and merges the Jun 16 remainder with the Jun 18 repo-research adopts.

## 0 · Why this exists

After the Jun 18 deep-fetch of ~71 repos (Section C of `REPO_RESEARCH_INDEX.md`), two pieces of system-level infrastructure showed up that fundamentally improve memory and observability/cost for the 18 GRID agents: **Mem0** (3-scope memory) and **Langfuse** (obs + cost + eval). These touch every agent — so they go FIRST, before per-agent expertise packs land on the new substrate. One painful migration, then 25 expertise packs. **caveman** is separate — it's a Claude-Code-session ergonomics skill (cuts MY chat tokens with Gaurav), NOT a production-cost lever for the 18 agents. It rides along as a low-risk dev-ergonomics install.

Picsart Agents was reviewed Jun 18; it's image-editing-only, validates our approval gate, and adds **one** backlog item: WhatsApp/Telegram as future CoS channels (post-launch). No structural change to this plan.

## 1 · Phase 1 — Foundation upgrades (1–2 days · \$0)

These touch every agent; do them first.

| #  | Item                                                                        | What it does                                                                                                                                                                                                  | Source                             |
|----|-----------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------|
| 1a | Wire Mem0 as 3-scope memory backbone                                        | Brand / Personal / Account scopes; migrate scattered files ( <code>brand_profile.json</code> , <code>voice_profile.json</code> , <code>performance_history.json</code> , <code>agent_learnings.jsonl</code> ) | <code>mem0ai/mem0</code>           |
| 1b | Wire Langfuse (self-hosted)                                                 | Traces + per-agent cost + LLM-as-judge eval; retire hand-built <code>cost_reporter</code> and most of the eval harness — gives REAL cost data before Phase 4 policy change so we can measure the win          | <code>langfuse/langfuse</code>     |
| 1c | Fix <code>rate_limit</code> decorator-order bug                             | Open chip from Jun 16 — rate limits silently inactive across <code>routes/*.py</code>                                                                                                                         | internal                           |
| 1d | AgentShield baseline scan                                                   | Run once, log findings, defer fixes to pre-client gate                                                                                                                                                        | <code>affaan-m/agentshield</code>  |
| 1e | caveman skill install (Claude-Code session ergonomics, NOT production cost) | Clone → inspect <code>install.sh</code> → install. Cuts MY chat-token use with Gaurav during the build. Does NOT touch the 18 agents' API calls.                                                              | <code>JuliusBrussee/caveman</code> |

**Exit criteria:** all 18 agents log to Langfuse, read/write memory via Mem0. Hand-built `cost_reporter` retired. Rate-limit middleware actually active. caveman live in this Claude Code session (separate concern, low-risk).

## 2 · Phase 2 — Data & research hardening (1 day · \$0)

| #  | Item                                | What it does                                                        | Source                                               |
|----|-------------------------------------|---------------------------------------------------------------------|------------------------------------------------------|
| 2a | Scraping alongside Apify            | Anti-bot resilience for trend-researcher + strategy-agent           | <code>D4Vinci/Scrapling</code>                       |
| 2b | supadata.ai for video → transcript  | Replaces Whisper pipeline in trend-researcher (key-gated)           | supadata.ai                                          |
| 2c | irinabuht12 google-meta-ads-ga4 MCP | Live metrics for data-analyst — closes the META_GRAPH_API_TOKEN gap | <code>irinabuht12-oss/google-meta-ads-ga4-mcp</code> |

**Exit criteria:** trend-researcher survives 5 consecutive scrapes without bot-block; data-analyst returns real GA4/Meta/Google-Ads metrics for askgauravai.

## 3 · Phase 3 — Expertise packs, one agent at a time (2–3 days · \$0)

Per Gaurav's "feed each agent hands-on" rule. **3a is the template** — finish it, then mass-produce.

| #  | Agent                         | Pack source(s)                                                                                                                                                                       | Notes                                  |
|----|-------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------|
| 3a | ad-strategist                 | <a href="#">AgriciDaniel/claude-ads</a> (250+ audit checks + creative pipeline)                                                                                                      | Template pack — finish first.          |
| 3b | seo-aeo-agent                 | <a href="#">AgriciDaniel/claude-seo</a> (25 sub-skills/18 agents) + <a href="#">zubair-trabzada/geo-seo-claude</a> (foundation) + <a href="#">aaron-he-zhu/seo-geo-claude-skills</a> | Mirror the 25-subskill fleet structure |
| 3c | creative-director             | <a href="#">AgriciDaniel/banana-claude</a> + <a href="#">ZeroLu/awesome-nanobanana-pro</a> prompt library                                                                            | Higgsfield engine stays gated on PRO   |
| 3d | strategy-agent                | <a href="#">phuryn/pm-skills</a> + <a href="#">minhvn0807/ai-business-skills</a> + <a href="#">AICMO/AiCMO-Marketing-Prompt-Collection</a>                                           | Discovery + launch + CMO frames        |
| 3e | brand-guardian                | <a href="#">arnabbagxd/Brand-building-skills</a>                                                                                                                                     | Naming + voice + positioning           |
| 3f | content-planner               | <a href="#">coreyhaines31/marketingskills</a> + <a href="#">OpenClaudia/openclaudia-skills</a>                                                                                       | Calendar skills                        |
| 3g | script-writer                 | <a href="#">AgriciDaniel/claude-blog</a> (5-gate 90/100 quality contract)                                                                                                            | Applies our AutoResearch loop          |
| 3h | carousel-designer             | <a href="#">banana-claude</a> + nano-banana prompts (shared with CD)                                                                                                                 | Image consistency                      |
| 3i | data-analyst                  | <a href="#">cognyai/claude-code-marketing-skills</a> + <a href="#">Zafer-Liu/Data-Analysis-Agent</a> (NL→SQL pattern)                                                                | Live-data hooks                        |
| 3j | funnel-specialist             | <a href="#">coreyhaines31/marketingskills</a> CRO subset                                                                                                                             |                                        |
| 3k | trend-researcher              | <a href="#">Affitor/affiliate-skills</a> + <a href="#">ruvnet/guardrail</a> (sentiment math)                                                                                         | Trend-scouting patterns                |
| 3l | seo subagents                 | already mostly covered by 3b                                                                                                                                                         |                                        |
| 3m | email-marketing-agent         | <a href="#">OpenClaudia/openclaudia-skills</a> (email subset) + Klaviyo REST direct                                                                                                  |                                        |
| 3n | community-manager             | mostly LLM + policy; light pack                                                                                                                                                      | LiteLLM tier change in Phase 4         |
| 3o | Lead Generator (ex-DM-hunter) | <a href="#">kenny589/gtm-flywheel</a> (15 GTM frames) + <a href="#">omkarcloud/google-maps-scraper</a>                                                                               | Local-biz ICP                          |
| 3p | AI-setter                     | <a href="#">kaymen99/sales-outreach-automation-langgraph</a> + <a href="#">chaitanyya/sales</a> (ICP-scoring)                                                                        | Port flow to our SDK                   |

Pure-math agents (trend-sentinel, performance-tracker, cost-tracker) need NO expertise pack.

**Exit criteria:** each agent has a [program.md](#) derived from its pack source(s); first-run output passes provenance + brand-voice checks.

## 4 • Phase 4 — LiteLLM policy + new capabilities (½ day)

| #  | Item                                          | What it does                                                                                                | Source                                         |
|----|-----------------------------------------------|-------------------------------------------------------------------------------------------------------------|------------------------------------------------|
| 4a | LiteLLM routing change for high-volume grunt  | Haiku 4.5 / Gemini 2.5 Flash / GPT-5 Mini for community-manager, Lead Generator, AI-setter → 5–12× cost cut | <a href="#">agents/_lib/model_gateway.py</a>   |
| 4b | google-maps-scraper wired into Lead Generator | Local-biz ICP harvesting                                                                                    | <a href="#">omkarcloud/google-maps-scraper</a> |
| 4c | google-ads-python client added (gated)        | Fires only when paid budgets confirmed per brand                                                            | <a href="#">googleads/google-ads-python</a>    |

**Exit criteria:** Langfuse shows the 3 high-volume agents on cheaper tiers; weekly cost projection drops ≥40%.

## 5 • Phase 5 — Validation gate (½ day)

| #  | Item                                                 | What it does                                    |
|----|------------------------------------------------------|-------------------------------------------------|
| 5a | F4 magnet copy via <a href="#">funnel_specialist</a> | Closes the F4 funnel item (parked since Jun 16) |

| #  | Item                               | What it does                                                  |
|----|------------------------------------|---------------------------------------------------------------|
| 5b | One supervised paid validation run | <code>GRID_PAID_OPS=1</code> , \$5/day cap on, watch Langfuse |
| 5c | End-to-end smoke test              | askgauravai full pipeline trace, golden-path + 2 edges        |

Exit criteria of Phase 5 = exit criteria of Master-Sequence Step 1 (Backend). Then move to Step 2 (Lovable front-end).

### 6 · Deferred (do NOT do in this plan)

- **Higgsfield PRO** + per-agent creative work (Master-Sequence Step 5)
- **Front-end (Lovable/v0)** (Master-Sequence Step 2)
- **ManyChat F3 / AI Setter live** — needs live offer + 2 links (purchase + Calendly)
- **Open Design mining** → `CREATIVE_PROMPTING_NOTES.md` (post-Phase-3 nice-to-have)
- **WhatsApp / Telegram CoS channels** — Picsart-inspired backlog item, post-launch
- **Postiz on Hetzner CX22** — Phase M, post-launch

### Phase-1 caveat (Jun 18, post-verification)

Caveman is Claude-Code-session terseness — it reduces MY tokens in the build chat with Gaurav, not the 18 agents' production costs. Real production cost wins come from Phase 4a (LiteLLM Haiku/Flash policy on high-volume agents) and from observability via Langfuse. Caveman = dev-ergonomics, demoted to 1e and explicitly scoped.

### 7 · Backlog carried from Jun 16 (status check)

| Item                                 | Status                                                                       |
|--------------------------------------|------------------------------------------------------------------------------|
| Apply 006 migration to prod          | ✅ DONE (Jun 16 part 2)                                                       |
| Scheduler worker deployed to Railway | ✅ DONE (Jun 16 part 2)                                                       |
| Phase J concierge router             | ✅ DONE                                                                       |
| K3 revision cap + scope-creep        | ✅ DONE                                                                       |
| YouTube OAuth permanent              | ✅ DONE                                                                       |
| F4 magnet copy                       | ⌚ → Phase 5a                                                                 |
| Supervised paid validation run       | ⌚ → Phase 5b                                                                 |
| rate_limit decorator-order bug       | ⌚ → Phase 1d                                                                 |
| Wave-3 hardening                     | ⌚ folded into Phase 1–4 (caveman + Langfuse + AgentShield together cover it) |
| Open Design mining                   | ⌚ deferred — Section 6                                                       |

### 8 · Picsart synthesis (1-paragraph reference)

Picsart Agents (TechCrunch Mar 16 2026): 4 agents — Flair (Shopify), Resize Pro, Remix, Swap. Image-editing only. WhatsApp + Telegram access. "Autonomy levels" = approval gate. Free + ~\$10/mo. **Wedge they own:** Shopify integration + Picsart image library. **Wedge we own:** 18-agent full-funnel marketing OS (strategy → content → publish → measure → learn). **Borrow:** WA/TG as future CoS channels (post-launch). **Tagline for the freebee:** "Picsart = 4 image agents. GRID = full marketing department on autopilot."

### 9 · Verification points (already confirmed by Gaurav Jun 18)

1. ✅ Order is correct: Foundation → Data → Expertise → Policy → Validation.
2. ✅ Higgsfield-PRO / Open-Design / ManyChat / FE stay deferred.
3. ✅ Phase-1 substrate migration BEFORE expertise packs (one painful migration, then 25 packs land on the new substrate).
4. ✅ Picsart triggers no structural change — only WA/TG backlog add.